MASTER DEVELOPMENT SPECIFICATION

MASTER ENGINEERING SPECIFICATION — V1.0

Status: FROZEN FOR V1 DEVELOPMENT

Version: 1.0.0

Market: Forex

Instruments: EURUSD, GBPUSD, USDJPY, USDCHF, USDCAD, AUDUSD, NZDUSD

Style: Intraday / Day Trading

Execution: EA

Primary Timeframes: H1 Context + M15 Execution

Architecture Principle: Deterministic Core → AI Intelligence Layer

---

1. SYSTEM OBJECTIVE

Sistem dirancang sebagai modular trading intelligence platform untuk menemukan dan mengeksekusi peluang intraday pada 7 major forex pairs.

Objective:

> Menghasilkan positive expectancy yang robust terhadap perubahan market regime, transaction cost, volatility, execution condition, dan perubahan kondisi pasar dengan risk yang terkontrol.

Primary optimization target **bukan**:

- maximum trade count;
- maximum win rate;
- maximum backtest profit;
- maximum Sharpe pada satu sample.

Primary objective:

```
ROBUST POSITIVE EXPECTANCY + CONTROLLED DRAWDOWN + REPRODUCIBILITY + EXECUTION ROBUSTN
ESS
```

---

2. CORE ARCHITECTURE

```
MARKET DATA
     ↓
DATA VALIDATOR
     ↓
FEATURE ENGINE
     ↓
REGIME ENGINE
     ↓
STRATEGY ENGINE
     ↓
CANDIDATE SIGNAL
     ↓
STRATEGY GUARDIAN
     ↓
AI INTELLIGENCE [V2]
     ↓
RISK ENGINE
     ↓
PORTFOLIO ENGINE
     ↓
EXECUTION SAFETY
     ↓
EXECUTION ENGINE
     ↓
POSITION MANAGER
     ↓
TRADE RESULT
     ↓
TRADE LOGGER
     ↓
ANALYTICS
     ↓
RESEARCH
```

AI adalah **optional layer**, bukan dependency dari deterministic core.

---

3. VERSION PRINCIPLE

V1

```
DATA → FEATURES → REGIME → STRATEGY → RISK → PORTFOLIO → EXECUTION
```

Tidak membutuhkan AI.

V2

```
V1 + AI Intelligence
```

V3

```
V2 + Advanced Intelligence
```

Tidak ada V2/V3 yang boleh mengubah hard safety authority.

---

4. AUTHORITY HIERARCHY

Urutan authority:

```
1. HARD SAFETY
2. EXECUTION SAFETY
3. RISK ENGINE
4. PORTFOLIO ENGINE
5. STRATEGY GUARDIAN
6. STRATEGY ENGINE
7. AI INTELLIGENCE
```

Lower-level component tidak boleh override higher-level constraint.

Contoh:

```
AI Confidence = 99%
```

tidak dapat override:

```
MAX_RISK_PER_TRADE = 0.50%
```

---

5. INSTRUMENTS

V1:

```
EURUSD
GBPUSD
USDJPY
USDCHF
USDCAD
AUDUSD
NZDUSD
```

Satu generalized strategy framework digunakan untuk seluruh pair.

Pair-specific parameter hanya boleh diperkenalkan jika:

```
Statistical evidence + OOS improvement + Walk-forward stability + Robustness + No mate
rial overfitting
```

Default:

> Generalized parameters first.

---

6. TIMEFRAME CONTRACT

H1

H1 digunakan untuk:

```
Market regime
Trend direction
Trend strength
Volatility context
EMA alignment
EMA slope
```

M15

M15 digunakan untuk:

```
Market structure
Range
Breakout
Retest
Pullback
Entry confirmation
SL
TP
Trailing
```

V1 tidak menggunakan M5.

---

7. CANDLE TIMING CONTRACT

Ini sekarang **mandatory**.

Strategy decision hanya dibuat berdasarkan **closed candle**.

Misalnya:

```
M15 candle 10:15–10:30
```

maka strategy baru boleh menggunakan candle tersebut setelah:

```
10:30
```

Decision timestamp:

```
T = candle close
```

Tidak boleh menggunakan data dari candle yang masih berjalan untuk entry decision.

Tick data masih boleh digunakan untuk:

- execution;
- spread monitoring;
- position management;
- SL/TP;

- broker state.

Tetapi bukan untuk menciptakan informasi future bagi strategy signal.

---

8. DATA ENGINE

Data Engine menyediakan:

```
Timestamp
Open
High
Low
Close
Bid
Ask
Spread
Volume jika tersedia
```

Data harus:

```
Timestamp consistent
Duplicate checked
Gap checked
OHLC validated
Broker source identified
Timezone normalized
```

Semua live input harus dapat direproduksi semaksimal mungkin dalam research/backtest.

---

9. DATA QUALITY STATES

Data Validator memiliki state:

```
VALID
WARNING
INVALID
```

INVALID

Contoh:

```
Missing required candle
Invalid OHLC
Timestamp corruption
Duplicate conflicting candle
Impossible price relationship
Required timeframe unavailable
```

Jika data `INVALID`:

```
NO NEW TRADE
```

---

10. FEATURE ENGINE

Feature Engine adalah:

> **Single Source of Truth untuk seluruh market feature.**

Tidak boleh ada:

```
Strategy menghitung ATR sendiri
Guardian menghitung ATR sendiri
AI menghitung ATR sendiri
Risk menghitung ATR sendiri
```

Semua mengambil feature dari Feature Engine.

---

11. FEATURE SNAPSHOT

Setiap decision memiliki immutable:

```
FeatureSnapshot
```

Minimal:

```
timestamp
symbol
```

```
timeframe
feature_version
```

dan seluruh feature yang diperlukan.

Snapshot harus dapat disimpan dan direproduksi.

---

12. H1 FEATURES

```
EMA50_H1
EMA200_H1
ADX14_H1
ATR14_H1
EMA50_Slope_H1
Close_H1
DistanceToEMA50_H1
DistanceToEMA200_H1
```

Distance:

```
DistanceToEMA50 = (Close - EMA50) / ATR
```

Directional sign dipertahankan.

---

13. M15 FEATURES

```
EMA50_M15
ADX14_M15
ATR14_M15

BBUpper_M15
BBMiddle_M15
BBLower_M15
BBWidth_M15

CandleBody
CandleRange
BodyATRRatio

SwingHigh
SwingLow

RangeHigh
```

```
RangeLow
RangeWidth

MarketStructure
PullbackDepth
BreakoutDistance
RetestDistance

Spread
SpreadATRRatio

Session
VolatilityState
```

---

14. NORMALIZATION

Price-distance feature sebisa mungkin dinormalisasi terhadap ATR.

Contoh:

```
EMA distance / ATR
Range width / ATR
Candle body / ATR
Spread / ATR
EMA slope / ATR
Pullback depth / ATR
Breakout distance / ATR
Retest distance / ATR
```

Tujuan:

```
mengurangi dependency terhadap absolute pip scale.
```

---

15. FEATURE VERSIONING

Setiap feature snapshot memiliki:

```
FEATURE_V1.0
```

Jika formula feature berubah:

```
FEATURE_V1.1
```

Dataset lama tidak boleh diam-diam dianggap menggunakan formula baru.

---

16. SWING DETECTION

Swing definition sekarang dikunci sebagai **confirmed structure only**.

Swing tidak boleh menggunakan future candle secara ilegal.

Sebuah swing baru dianggap confirmed setelah jumlah candle konfirmasi yang telah ditentukan terpenuhi.

Parameter:

```
SWING_LEFT_BARS
SWING_RIGHT_BARS
SWING_MIN_DISTANCE
SWING_MIN_SIGNIFICANCE
```

semuanya configuration/experimental parameter.

Critical rule:

> Confirmation delay harus diperhitungkan dalam backtest.

Swing yang baru diketahui setelah candle T+N tidak boleh dianggap sudah diketahui pada T.

---

17. MARKET STRUCTURE

Market structure M15 terdiri dari:

```
HH = Higher High
HL = Higher Low
LH = Lower High
LL = Lower Low
```

Bullish structure:

```
HH + HL
```

Bearish:

```
LH + LL
```

Structure belum valid jika swing belum confirmed.

---

18. REGIME ENGINE

Regime state:

```
UNKNOWN
SIDEWAYS
TRANSITION
TREND
TREND_EXHAUSTION
```

Regime Engine menghasilkan:

```
current_state
previous_state
state_timestamp
transition_reason
score
confidence
regime_version
```

---

19. REGIME PERSISTENCE

Regime tidak boleh berubah hanya karena satu candle tanpa confirmation.
Configuration:

```
REGIME_CONFIRMATION_BARS
```

State transition harus melalui deterministic rule.

Contoh:

```
SIDEWAYS
↓
breakout detected
↓
TRANSITION
↓
trend confirmation
↓
TREND
```

---

20. SIDEWAYS DETECTION

SIDEWAYS tidak ditentukan hanya oleh ADX.

Baseline menggunakan:

```
ADX
EMA slope
Volatility compression
Range structure
```

ADX baseline:

```
ADX_H1 < 20
```

Tetapi bukan automatic decision.

---

21. EMA SLOPE

Slope harus deterministic.

Baseline:

```
EMA50_Slope = (EMA50[t] - EMA50[t-N]) / ATR[t]
```

Parameter:

```
EMA_SLOPE_LOOKBACK
```

harus configurable.

---

22. VOLATILITY COMPRESSION

Baseline menggunakan:

```
BBWidth
ATR relative state
```

Contoh normalized BB width:

```
BBWidth / ATR
```

Threshold merupakan experimental parameter.

---

23. SIDEWAYS SCORE

Baseline:

```
SidewaysScore = ADX condition + EMA slope condition + volatility condition  + range co
ndition
```

Regime:

```
Score >= SIDEWAYS_MIN_SCORE
```

baru dapat menjadi SIDEWAYS.

Threshold tidak dianggap optimal sebelum validation.

---

24. RANGE DETECTION

Baseline:

```
RANGE_LOOKBACK = 20 M15 candles
```

Range boundary berasal dari confirmed structure/high-low yang valid.

```
RangeHigh
RangeLow
RangeWidth
```

Range harus memiliki minimum structural quality.

---

25. RANGE INTERACTION

Interaction tolerance berbasis ATR:

```
ABS(price - boundary) <= RANGE_INTERACTION_TOLERANCE × ATR
```

Tolerance adalah configurable parameter.

---

26. RANGE VALIDITY

Baseline:

```
≥ 2 upper interactions
≥ 2 lower interactions
```

Namun interaction harus berasal dari distinct price events, bukan repeated counting terhadap candle yang sama.

Range harus:

```
Width > minimum
Width < maximum
```

dengan threshold configurable.

---

27. RANGE ZONE

Outer zone baseline:

```
20%
```

BUY zone:

```
Lower 20% of range
```

SELL zone:

```
Upper 20% of range
```

Middle range:

```
NO ENTRY
```

---

28. REJECTION — BULLISH

Baseline:

```
Low interacts with RangeLow + Close > Open + Close berada pada upper portion candle
```

Upper-close ratio:

```
(Close - Low) / (High - Low)
```

minimum ratio menjadi configurable.

---

29. REJECTION — BEARISH

Baseline:

```
High interacts with RangeHigh + Close < Open + Close berada pada lower portion candle
```

Lower-close ratio:

```
(High - Close) / (High - Low)
```

minimum ratio configurable.

---

30. RANGE BUY

```
REGIME = SIDEWAYS + Price in BUY_ZONE + RangeLow interaction + Bullish rejection + Can
dle closes back inside range
```

Entry:

```
confirmation candle close
```

---

31. RANGE SELL

```
REGIME = SIDEWAYS + Price in SELL_ZONE + RangeHigh interaction + Bearish rejection + C
andle closes back inside range
```

Entry:

```
confirmation candle close
```

---

32. RANGE STOP LOSS

BUY:

```
SL = structural invalidation or RangeLow whichever provides the required invalidation
```

```
- ATR × SL_BUFFER
```

SELL:

```
SL = structural invalidation or RangeHigh +  ATR × SL_BUFFER
```

Baseline:

```
SL_BUFFER = 0.2 ATR
```

SL must remain outside invalidation structure.

---

33. RANGE TAKE PROFIT

Supported modes:

```
MID_RANGE
OPPOSITE_BOUNDARY
FIXED_R
HYBRID
```

Baseline:

```
TP1 = Mid Range
TP2 = Opposite Boundary
```

TP mode must be configuration-controlled.

---

34. BREAKOUT DEFINITION

Bullish:

```
Close > RangeHigh
```

Bearish:

```
Close < RangeLow
```

Only closed candle breakout is valid.

Wick-only:

```
INVALID
```

---

35. BREAKOUT QUALITY

Components:

```
Structure
Candle Strength
Volatility Expansion
ADX Expansion
```

Baseline candle strength:

```
Body >= 0.5 × ATR
```

Volatility:

```
ATR_current > ATR_reference
```

ADX:

```
ADX_current > ADX_reference
```

All thresholds configurable.

---

36. TRANSITION STATE

Valid breakout:

```
SIDEWAYS
↓
TRANSITION
```

TRANSITION berarti:

> breakout sedang dievaluasi untuk menentukan apakah berkembang menjadi trend atau gagal.

TRANSITION bukan TREND.

---

37. BREAKOUT RETEST

Bullish:

```
Breakout
↓
Retrace
↓
Former RangeHigh interaction
↓
Bullish confirmation
↓
BUY
```

Bearish:

```
Breakdown
↓
Retrace
↓
Former RangeLow interaction
↓
Bearish confirmation
↓
SELL
```

Tolerance:

```
ATR normalized
```

---

38. BREAKOUT FAILURE

Jika:

```
Breakout
↓
Price kembali masuk range
```

maka:

```
BREAKOUT_FAILED
```

Transition:

```
TRANSITION
↓
SIDEWAYS
```

jika range masih valid.

Jika range invalid:

```
TRANSITION
↓
UNKNOWN
```

---

39. TREND BULLISH

H1:

```
Close > EMA200
EMA50 > EMA200
EMA50 slope > threshold
ADX >= TREND_ADX
```

M15:

```
HH + HL
```

---

40. TREND BEARISH

H1:

```
Close < EMA200
EMA50 < EMA200
EMA50 slope < -threshold
ADX >= TREND_ADX
```

M15:

```
LH + LL
```

---

41. TREND EXHAUSTION

TREND_EXHAUSTION digunakan ketika trend belum sepenuhnya invalid tetapi kualitas trend menurun.

Contoh signals:

```
ADX deterioration
EMA slope deterioration
Structure deterioration
Volatility exhaustion
```

Tidak boleh langsung dianggap reversal.

Default action:

```
No new aggressive trend entry
```

Existing positions mengikuti Position Manager.

---

42. TREND PULLBACK

Flow:

```
TREND
↓
IMPULSE
↓
PULLBACK
↓
PULLBACK ZONE
↓
STRUCTURE / EMA INTERACTION
↓
CONFIRMATION
↓
ENTRY
```

Tidak cukup hanya:

```
Price > EMA
```

atau:

```
Price < EMA
```

---

43. TREND BUY

```
Bullish H1 trend + Bullish M15 structure + Valid pullback + Pullback reaches valid zon
e + Bullish confirmation
```

Baseline:

```
Close > Previous Candle High
```

---

44. TREND SELL

```
Bearish H1 trend + Bearish M15 structure + Valid pullback + Pullback reaches valid zon
e + Bearish confirmation
```

Baseline:

```
Close < Previous Candle Low
```

---

45. TREND STOP LOSS

BUY:

```
Pullback SwingLow - ATR × buffer
```

SELL:

```
Pullback SwingHigh + ATR × buffer
```

Structure invalidation memiliki priority lebih tinggi daripada arbitrary distance.

---

46. STRATEGY OUTPUT CONTRACT

Strategy Engine **tidak boleh execute trade**.

Output:

```
CandidateSignal
```

Minimal:

```
trade_id
symbol
timestamp
direction
strategy
setup
entry_reference
stop_loss
```

```
take_profit_plan
regime
feature_version
strategy_version
reason_codes
```

Candidate dapat:

```
BUY
SELL
NO_SIGNAL
```

---

47. STRATEGY GUARDIAN

Guardian melakukan deterministic validation.

Output:

```
APPROVED
```

atau:

```
REJECTED
```

Dengan reason code.

Contoh:

```
TREND_VALID
STRUCTURE_VALID
PULLBACK_VALID
ENTRY_ZONE_VALID
```

Guardian tidak mengubah strategy rules.

---

48. AI LAYER

AI hanya menerima:

```
ApprovedCandidate + FeatureSnapshot
```

AI tidak mencari trade dari raw market data pada V2.

Pipeline:

```
Market
↓
Feature
↓
Deterministic Strategy
↓
Candidate
↓
Guardian
↓
AI
```

---

49. AI OUTPUT CONTRACT

AI output wajib structured.

```
P_REACH_1R
P_REACH_2R
P_HIT_SL
EXPECTED_R
CONFIDENCE
MODEL_VERSION
```

Tidak boleh output seperti:

```
BUY because market looks strong.
```

---

50. AI MATHEMATICAL CONTRACT

Probability harus berada:

```
0 <= P <= 1
```

Expected R harus dapat direkonstruksi dari model/contract yang digunakan.

Invalid:

```
NaN
missing probability
probability > 1
probability < 0
schema mismatch
```

→ `AI_ERROR`.

---

51. AI FAILURE POLICY

Jika:

```
Timeout
Unavailable
Invalid response
Schema mismatch
Missing feature
Model loading failure
```

default:

```
REJECT TRADE
```

Fallback hanya boleh digunakan jika secara eksplisit diuji dan disetujui.

---

52. RISK ENGINE

Baseline:

```
BASE_RISK = 0.50%
```

Position size:

```
Risk Amount ÷ Monetary Loss Per Lot at Actual SL
```

Risk dihitung berdasarkan:

```
actual entry
actual SL
actual contract specification
actual tick/pip value
```

Bukan fixed lot.

---

53. RISK HARD LIMIT

```
MAX_RISK_PER_TRADE = 0.50%
```

Tidak dapat diubah oleh:

```
AI
Strategy
Signal Publisher
Customer system
```

---

54. LOSING STREAK

Baseline experimental multiplier:

Consecutive Loss	Risk Multiplier
0–1	100%
2	80%
3	60%
4	40%
5	20%
≥6	Review/Pause

Multiplier tidak boleh meningkatkan risk di atas base risk.

---

55. DRAWDOWN CONTROL

Gunakan:

```
High Water Mark
```

Baseline:

Drawdown	Multiplier
0–2%	100%
2–4%	75%
4–6%	50%
6–8%	25%
>8%	PAUSE

Final:

```
FinalRisk = BaseRisk × LosingStreakMultiplier × DrawdownMultiplier
```

Gunakan multiplier paling konservatif.

---

56. PORTFOLIO ENGINE

Portfolio Engine menghitung:

```
Open Positions
Pending Orders
Pair Exposure
Currency Exposure
Correlation
Aggregate Risk
```

Portfolio Engine memiliki authority untuk reject trade.

---

57. CURRENCY EXPOSURE

Exposure dihitung berdasarkan currency legs.

Contoh:

```
EURUSD BUY
EUR +1
USD -1

GBPUSD BUY
GBP +1
USD -1
```

Aggregate:

```
USD = -2
```

Portfolio engine harus memperhitungkan exposure gabungan.

---

58. PORTFOLIO HARD LIMITS

Configuration:

```
MAX_PORTFOLIO_RISK
MAX_CURRENCY_EXPOSURE
MAX_CORRELATED_EXPOSURE
MAX_POSITIONS
```

Nilai baseline final ditentukan melalui research.

---

59. CORRELATION

Gunakan rolling correlation sebagai portfolio intelligence.

Correlation:

```
INFORMATION
```

bukan satu-satunya trading signal.

Hard constraints dapat menggunakan correlation apabila telah tervalidasi.

---

60. SPREAD

Tidak menggunakan fixed universal:

```
Spread < 2 pips
```

Gunakan:

```
Spread / ATR
```

dan:

```
Absolute Spread
```

Jika threshold terlampaui:

```
NO NEW TRADE
```

Existing positions tidak otomatis ditutup hanya karena spread filter.

---

61. SLIPPAGE

Backtest minimal:

```
NORMAL
MODERATE
STRESS
```

Spread scenarios:

```
1.0 ×
1.5 ×
2.0 ×
```

Slippage juga disimulasikan.

---

62. SESSION ENGINE

Session:

```
ASIA
LONDON
NEW_YORK
LONDON_NY_OVERLAP
```

Timezone harus didefinisikan secara eksplisit dan konsisten.

Session restriction tidak boleh diterapkan hanya berdasarkan asumsi.

Performance harus dicatat per session terlebih dahulu.

---

63. NEWS TESTING

Dua baseline experiment:

```
MODE A
No News Filter

MODE B
High Impact News Filter
```

Compare:

```
PF
Expectancy
DD
Trade Count
Missed Winners
Execution Quality
```

Tidak ada asumsi bahwa news filter otomatis lebih baik.

---

64. ANOMALY ENGINE

Mendeteksi:

```
Spread Spike
ATR Explosion
Feed Abnormality
Execution Abnormality
```

```
Extreme Volatility
Broker Abnormal Response
```

Hard anomaly:

```
NO NEW TRADE
```

Existing position:

```
Hard Exit + Position Safety Management
```

sesuai state yang tersedia.

---

65. GLOBAL STATE MACHINE

Trading lifecycle:

```
INITIALIZING
READY
SCANNING
CANDIDATE_FOUND
VALIDATING
RISK_CHECK
EXECUTING
MANAGING
PAUSED
EMERGENCY
```

Regime state terpisah:

```
UNKNOWN
SIDEWAYS
TRANSITION
TREND
TREND_EXHAUSTION
```

Trading state ≠ market regime.

---

66. EVENT MODEL

Primary analysis event:

```
NEW_M15_CANDLE
```

Pipeline:

```
NEW_M15_CANDLE
↓
DATA_VALIDATION
↓
FEATURE_UPDATE
↓
REGIME_UPDATE
↓
PORTFOLIO_UPDATE
↓
ANOMALY_CHECK
↓
SCAN
```

Tick-level event digunakan untuk:

```
execution
position management
broker synchronization
spread monitoring
```

---

67. MASTER DECISION FLOW

```
NEW M15 CLOSED CANDLE
          ↓
DATA VALIDATION
          ↓
FEATURE UPDATE
          ↓
REGIME UPDATE
          ↓
PORTFOLIO UPDATE
          ↓
ANOMALY CHECK
          ↓
SCAN 7 PAIRS
          ↓
CANDIDATE?
```

```
      ├── NO → WAIT
      ↓
STRATEGY GUARDIAN
      ├── REJECT → LOG
      ↓
AI [V2]
      ├── ERROR → REJECT
      ↓
RISK ENGINE
      ├── REJECT → LOG
      ↓
PORTFOLIO ENGINE
      ├── REJECT → LOG
      ↓
EXECUTION SAFETY
      ├── REJECT → LOG
      ↓
EXECUTION ENGINE
      ↓
POSITION MANAGEMENT
      ↓
EXIT
      ↓
TRADE LOGGER
      ↓
PERFORMANCE ANALYSIS
```

---

68. EXECUTION ENGINE

Execution Engine deterministic.

Responsibilities:

```
Calculate Lot
Validate Order
Submit
Modify
SL
TP
Partial Close
Trailing
Retry
Broker Error Handling
Position Synchronization
```

AI tidak memiliki broker API access.

---

69. EXECUTION SAFETY

Sebelum order:

```
Symbol valid
Market open
Price valid
Spread valid
SL valid
TP valid
Lot valid
Margin sufficient
Risk valid
Portfolio valid
Broker constraints valid
```

Jika gagal:

```
NO ORDER
```

---

70. POSITION MANAGEMENT

Position Manager menangani:

```
SL
TP
Partial Close
Break-even
Trailing
Structure deterioration
Trend failure
```

Hard exit selalu memiliki authority tertinggi.

---

71. PROFIT MANAGEMENT

Baseline trend:

```
50% → 1R
25% → 2R
```

```
25% → Runner
```

Runner:

```
BUY → Higher Low trailing
SELL → Lower High trailing
```

ATR trailing adalah experimental alternative.

---

72. BREAK-EVEN

Tidak:

```
+0.5R → automatic BE
```

Baseline:

```
Profit milestone reached + new protective structure formed + trade protection conditio
n valid
```

baru BE/profit lock diperbolehkan.

---

73. TRADE ID

Format:

```
TRD-YYYYMMDD-SYMBOL-SEQ
```

Contoh:

```
TRD-20260826-EURUSD-001
```

Trade ID harus immutable.

---

74. TRADE EVENT IDENTITY

Trade ID menghubungkan:

```
Candidate
→ Guardian
→ AI Decision
→ Risk Decision
→ Portfolio Decision
→ Order
→ Position
→ Exit
→ Result
```

---

75. TRADE LOG

Minimal:

```
Timestamp
Trade ID
Symbol
Direction
Session
Regime
Strategy
Setup

Entry
SL
TP1
TP2

Risk %
Position Size

Spread
ATR
ADX

Feature Version
Strategy Version
AI Model Version

Guardian Result
AI Result
Risk Result
Portfolio Result
Execution Result
```

```
Exit
R Multiple
MAE
MFE

Final Result
```

---

76. DECISION AUDIT

Setiap trade harus dapat menjawab:

```
WHY WAS THIS TRADE TAKEN?
```

Contoh:

```
Regime: TREND
Strategy: TREND_PULLBACK
Guardian: APPROVED
AI: P(+1R)=74%
Risk: 0.35%
Portfolio: APPROVED
Execution: APPROVED
```

Loss tidak berarti decision salah.

Audit menjelaskan:

> apakah trade memenuhi rule saat decision dibuat.

---

77. IMMUTABLE DECISION SNAPSHOT

Saat trade approved, simpan:

```
Market Snapshot
Feature Snapshot
Regime Snapshot
Strategy Decision
Guardian Decision
AI Decision
Risk Decision
```

```
Portfolio Decision
Execution Decision
```

Tidak boleh ditimpa setelah trade selesai.

---

78. ANALYTICS

Metrics:

```
Profit Factor
Expectancy
Win Rate
Average Win
Average Loss
Average R
Max Drawdown
Recovery Factor
Sharpe
Sortino
Trade Count
Longest Losing Streak
Longest Winning Streak
MAE
MFE
```

Breakdown:

```
Pair
Session
Regime
Strategy
Direction
Time
AI Confidence
Model Version
```

---

79. EXPECTANCY

Primary metric:

```
Expectancy = Σ Trade R / Number of Trades
```

atau equivalent formulation berdasarkan average win/loss.

Semua comparison V1/V2 harus menggunakan definisi metric yang sama.

---

80. MODEL MONITORING

Monitor:

```
Expected Performance vs Live Performance
```

Jika deterioration signifikan:

```
MODEL_DEGRADATION_ALERT
```

Alert tidak otomatis:

```
change model
```

atau:

```
change strategy
```

---

81. RESEARCH ENGINE

Research sepenuhnya terpisah dari live execution.

```
LIVE DATA
↓
DATASET
↓
HYPOTHESIS
↓
BACKTEST
↓
OOS
↓
WALK FORWARD
↓
APPROVAL
```

```
↓
DEPLOYMENT
```

Research tidak boleh langsung mengubah production configuration.

---

82. LIVE SELF-LEARNING

V1/V2 awal:

```
DISABLED
```

Dilarang:

```
Live Loss
↓
Auto Retrain
↓
Auto Deploy
```

Allowed:

```
Live Data
↓
Offline Dataset
↓
Training
↓
Validation
↓
Approval
↓
Deployment
```

---

83. CUSTOMER INSIGHT

Customer analytics terisolasi dari trading decision.

Menganalisis:

```
Signal Frequency
Retention
```

```
Churn
Signal Timing
Signal Clarity
Drawdown Tolerance
Customer Complaints
```

Tidak boleh:

```
Customer Complaint
↓
Trading Rule Changed
```

---

84. SIGNAL PUBLISHER

Canonical event:

```
TradeEvent
```

digunakan oleh:

```
Telegram
Discord
Website
Copy Trade
```

Semua channel menggunakan decision yang sama.

Tidak boleh channel membuat trade decision sendiri.

---

85. VERSIONING

Strategy:

```
STRATEGY_V1.0
```

Feature:

```
FEATURE_V1.0
```

Dataset:

```
DATASET_V1.0
```

AI:

```
AI_MODEL_V1.0
```

Deployment harus menyimpan:

```
Training Period
Dataset Version
Feature Version
Model Version
Strategy Version
Validation Result
Deployment Date
```

---

86. CONFIGURATION

Semua parameter berada di configuration layer.

Contoh:

```
EMA_FAST
EMA_SLOW

ADX_PERIOD
ATR_PERIOD

BB_PERIOD
BB_DEVIATION

RANGE_LOOKBACK
RANGE_ZONE
RANGE_INTERACTION_TOLERANCE

BREAKOUT_BODY_ATR
BREAKOUT_ATR_EXPANSION
```

```
TREND_ADX
SIDEWAYS_ADX

EMA_SLOPE_LOOKBACK
REGIME_CONFIRMATION_BARS

SL_BUFFER

BASE_RISK
MAX_RISK

DD_MULTIPLIER
LOSS_MULTIPLIER

MAX_PORTFOLIO_RISK
MAX_CURRENCY_EXPOSURE
MAX_CORRELATED_EXPOSURE
```

---

87. PARAMETER CLASSIFICATION

Setiap parameter memiliki status:

```
FIXED
CONFIGURABLE
EXPERIMENTAL
OPTIMIZED
```

Baseline:

> **bukan optimal parameter.**

Contoh:

```
ADX = 20
```

berarti:

```
BASELINE_VALUE
```

bukan:

```
OPTIMAL_VALUE
```

---

88. HIDDEN PARAMETER POLICY

Dilarang:

```
Magic number
Hidden threshold
Undocumented multiplier
Undocumented timeout
Undocumented filter
```

Setiap angka yang mempengaruhi decision harus:

```
configuration atau derived mathematically dari documented input.
```

---

89. ERROR STATES

```
DATA_ERROR
FEATURE_ERROR
REGIME_ERROR
STRATEGY_ERROR
AI_ERROR
RISK_ERROR
PORTFOLIO_ERROR
EXECUTION_ERROR
BROKER_ERROR
```

Setiap error:

```
Timestamped
Categorized
Logged
Traceable
```

Critical error:

```
Alert + NO NEW TRADE
```

---

90. HEALTH MONITOR

Monitor:

```
Data Feed
Feature Engine
EA State
Broker Connection
Execution
AI Availability
Risk Engine
Portfolio State
Logging
Signal Publisher
```

System harus dapat menjawab:

```
IS SYSTEM HEALTHY?
```

---

91. FAIL-SAFE

Jika critical component unavailable:

```
NO NEW TRADE
```

Contoh:

```
Market data invalid
Risk engine unavailable
Portfolio state unknown
Execution state inconsistent
```

Existing positions tetap ditangani oleh available hard-safety mechanisms.

---

92. BACKTEST DATA CONTRACT

Backtest harus menggunakan data yang sama secara semantik dengan live system.

Backtest harus mempertimbangkan:

```
Spread
Slippage
Commission jika berlaku
Swap jika relevan
Broker constraints
Execution assumptions
Candle timing
Swing confirmation delay
```

---

93. BASELINE BACKTEST

Stage 1:

```
Baseline Parameters
```

Tujuan:

> Membuktikan apakah konsep memiliki edge.

Tidak melakukan optimization sebelum baseline selesai.

---

94. SENSITIVITY TEST

Test parameter secara:

```
individual
atau
logical group
```

Tujuan:

```
Parameter Plateau
```

bukan:

```
Best Parameter
```

---

95. OVERFITTING CONTROL

Tidak boleh melakukan:

```
50+ optimized parameters
```

Optimization priority:

```
1. Regime
2. Entry
3. Exit
4. Risk
```

Parameter tanpa robustness harus dihapus.

---

96. OUT-OF-SAMPLE

Contoh:

```
2021–2024 = IS
2025 = OOS
```

OOS:

```
NEVER USED FOR OPTIMIZATION
```

---

97. WALK-FORWARD

Contoh:

```
Train 2021–2022
Test  2023

Train 2021–2023
Test  2024

Train 2021–2024
Test  2025
```

Tujuan:

> Menguji adaptasi terhadap perubahan market regime.

---

98. MONTE CARLO

Randomize:

```
Trade order
Winning sequence
Losing sequence
Slippage
```

Evaluate:

```
Potential DD
Potential Losing Streak
Equity Variation
Risk of Ruin
```

---

99. STRESS TEST

Spread:

```
1.0 ×
1.5 ×
2.0 ×
```

Slippage:

```
NORMAL
MODERATE
STRESS
```

Strategy hanya dianggap robust jika edge tidak bergantung pada friction assumption yang terlalu ideal.

---

100. REGIME ANALYSIS

Wajib:

```
SIDEWAYS
TRANSITION
TREND
TREND_EXHAUSTION
```

Metrics:

```
Trade Count
PF
Expectancy
Average R
DD
```

---

101. PAIR ANALYSIS

Wajib:

```
EURUSD
GBPUSD
USDJPY
USDCHF
USDCAD
AUDUSD
NZDUSD
```

Pair tidak boleh dihapus hanya karena satu periode buruk.

Removal membutuhkan:

```
Multi-period evidence + OOS evidence + Walk-forward evidence  + Research approval
```

---

102. SESSION ANALYSIS

Wajib:

```
Asia
London
New York
London/NY Overlap
```

Session restriction hanya boleh diterapkan berdasarkan evidence.

---

103. NEWS ANALYSIS

Bandingkan:

```
No News Filter vs News Filter
```

Minimal:

```
PF
Expectancy
DD
Trade Count
Missed Winners
Execution Quality
```

---

104. V1 PASS CRITERIA

Tidak ada angka universal yang dipaksakan.

V1 viable jika:

```
Positive Expectancy + PF > 1 + Acceptable Drawdown + Sufficient Trade Count + Multi-Ye
ar Stability + OOS Positive + Walk-Forward Acceptable + Monte Carlo Acceptable + Stres
```

```
s-Test Viable
```

"Acceptable" harus ditentukan sebelum final validation berdasarkan risk objective system.

---

105. V1 → V2 GATE

AI baru boleh dikembangkan setelah:

```
✓ Strategy logic stable
✓ Backtest reproducible
✓ No look-ahead bias
✓ OOS positive
✓ Walk-forward acceptable
✓ Monte Carlo acceptable
✓ Spread stress acceptable
✓ Slippage stress acceptable
✓ Multi-pair tested
✓ Multi-session tested
✓ Regime performance understood
✓ Complete trade logging
```

---

106. AI DATASET

Dataset row merepresentasikan:

```
FEATURES AT T + CANDIDATE AT T + DECISION CONTEXT AT T + FUTURE OUTCOME AS LABEL
```

Future information hanya boleh berada di:

```
LABEL
```

Tidak boleh berada di:

```
FEATURE
```

---

107. AI LABELS

Minimal:

```
Reached +1R
Reached +2R
Hit SL
MFE
MAE
Final R
```

Label horizon harus didefinisikan secara eksplisit.

---

108. AI DATA LEAKAGE RULE

Dilarang menggunakan:

```
Future High
Future Low
Future Close
Future Indicator
Future Spread
Future Position Outcome
Future MFE
Future MAE
```

sebagai feature.

---

109. V1 VS V2

V2 tidak dinilai hanya berdasarkan:

```
Higher Equity
```

Compare:

```
PF
Expectancy
Average R
Max DD
Recovery Factor
```

```
Sharpe
Sortino
Trade Count
Losing Streak
OOS
Walk Forward
Monte Carlo
Stress
```

---

110. AI ACCEPTANCE

AI diterima hanya jika improvement:

```
Robust + Repeatable + OOS supported + Walk-forward supported + Stress supported
```

bukan hanya:

```
better in-sample equity curve.
```

---

111. AI REJECTION

Contoh:

```
V1
PF = 1.35
DD = 8%
OOS = positive

V2
PF = 1.60
DD = 14%
OOS = negative
```

Result:

```
AI_REJECTED
```

Production candidate:

```
V1
```

---

112. V3

V3 hanya setelah V2 terbukti.

Potential modules:

```
AI Dynamic Exit
Anomaly Intelligence
Correlation Intelligence
Advanced Portfolio Optimization
Research Automation
```

Setiap component:

```
INDIVIDUAL EXPERIMENT
```

Tidak boleh memasukkan semuanya sekaligus.

---

113. SOFTWARE ARCHITECTURE

Logical architecture:

```
/EA

/Core
    Config
    StateMachine
    EventBus
    ErrorHandler
    Types

/Data
    MarketData
    DataValidator
    FeatureEngine
    SessionEngine

/Strategy
    RegimeEngine
    RangeStrategy
```

```
    BreakoutStrategy
    TrendStrategy
    StrategyGuardian

/Risk
    PositionSizing
    DrawdownControl
    LosingStreak
    PortfolioExposure
    CurrencyExposure

/Execution
    ExecutionSafety
    OrderManager
    PositionManager
    ExitManager

/AI
    Orchestrator
    MarketScanner
    ProbabilityAgent

/Signal
    TradeEvent
    SignalPublisher

/Analytics
    TradeLogger
    PerformanceAnalyzer
    ModelMonitor
    CustomerInsight

/Research
    DatasetBuilder
    ResearchEngine

/Monitoring
    HealthMonitor
    Alerts
    Audit
```

Catatan:

> Ini adalah logical architecture. Physical class/file decomposition dapat berkembang selama contract tetap sama.

---

114. COMPONENT CONTRACTS

Feature Engine

```
INPUT: MarketData
OUTPUT: FeatureSnapshot
```

Regime Engine

```
INPUT: FeatureSnapshot
OUTPUT: RegimeState
```

Strategy

```
INPUT: FeatureSnapshot, RegimeState
OUTPUT: CandidateSignal
```

Guardian

```
INPUT:
CandidateSignal
FeatureSnapshot
RegimeState

OUTPUT: GuardianDecision
```

AI

```
INPUT:
ApprovedCandidate
FeatureSnapshot

OUTPUT: AIProbability
```

Risk

```
INPUT:
Candidate
AIProbability
PortfolioState
```

```
OUTPUT: RiskDecision
```

Portfolio

```
INPUT:
RiskDecision
PortfolioState

OUTPUT: PortfolioDecision
```

Execution

```
INPUT:
ApprovedTradeDecision

OUTPUT: ExecutionResult
```

---

115. MODULE RESPONSIBILITY RULE

Setiap module harus mempunyai satu primary responsibility.

Contoh:

```
FeatureEngine
```

tidak boleh:

```
place order
```

Strategy:

```
generate candidate
```

bukan:

```
calculate final lot
```

Risk:

```
calculate risk
```

bukan:

```
send Telegram
```

Execution:

```
broker interaction
```

bukan:

```
AI prediction
```

---

116. TESTABILITY

Setiap critical module harus dapat diuji tanpa broker.

Minimal:

```
Feature tests
Regime tests
Strategy tests
Risk tests
Portfolio tests
Position sizing tests
Execution validation tests
```

Strategy logic harus dapat dijalankan terhadap historical fixture.

---

117. DETERMINISM REQUIREMENT

Untuk input yang sama:

```
Same Data + Same Configuration + Same Feature Version + Same Strategy Version
```

harus menghasilkan:

```
Same Decision
```

kecuali component memang secara eksplisit stochastic dan randomness-nya dikontrol/recorded.

V1 strategy:

> deterministic.

---

118. AUDITABILITY

Setiap decision harus mempunyai:

```
timestamp
symbol
feature_version
strategy_version
config_version
regime
candidate
guardian
risk
portfolio
execution
```

V2 menambahkan:

```
model_version
model_output
```

---

119. CONFIG VERSION

Selain:

```
STRATEGY_VERSION
```

kita juga wajib memiliki:

```
CONFIG_VERSION
```

Karena strategy version yang sama dengan parameter berbeda menghasilkan behavior berbeda.

Deployment identity menjadi:

```
Strategy Version + Feature Version + Config Version + Dataset Version + Model Version
```

---

120. DEPLOYMENT IMMUTABILITY

Live deployment tidak boleh mengubah parameter secara silent.

Perubahan harus:

```
New Config Version
```

dan dicatat:

```
Who
When
Why
Previous Version
New Version
Validation Reference
```

---

121. PRODUCTION SAFETY

Production system harus memiliki:

```
Trading Enabled/Disabled
Emergency Stop
```

```
Max Risk
Max Portfolio Risk
Broker Connection Status
Data Health
Execution Health
```

Emergency Stop memiliki authority tertinggi.

---

122. RESEARCH → PRODUCTION GATE

Tidak ada:

```
Research
↓
Direct Production
```

Flow:

```
Research
↓
Candidate Version
↓
Backtest
↓
OOS
↓
Walk Forward
↓
Stress
↓
Approval
↓
Release Version
↓
Forward Test
↓
Production
```

---

123. DEVELOPMENT PHASES

Phase 0 — Foundation

```
Configuration
Data Engine
Data Validator
Feature Engine
Feature Snapshot
State Machine
Logging
```

Phase 1 — Deterministic Strategy

```
Regime
Range
Breakout
Trend
Guardian
```

Phase 2 — Risk

```
Position Sizing
Losing Streak
Drawdown
Portfolio
Currency Exposure
Correlation
```

Phase 3 — Execution

```
Order Manager
Position Manager
Exit Manager
Execution Safety
Broker Synchronization
```

Phase 4 — Research

```
Backtest
Sensitivity
OOS
Walk Forward
```

```
Monte Carlo
Stress
```

Phase 5 — Forward Test

```
Broker
Spread
Slippage
Execution
Signal
Position Management
```

Phase 6 — AI Dataset

```
Feature Snapshot
Candidate
Outcome
```

Phase 7 — V2 AI

```
Probability
Guardian Enhancement
Orchestrator
```

Phase 8 — Production Intelligence

```
Signal Publisher
Analytics
Model Monitoring
Customer Insight
```

---

124. DEVELOPMENT PRIORITY

Urutan development:

```
1. Configuration
2. Core Types / Contracts
```

```
3. Data Engine
4. Data Validator
5. Feature Engine
6. Feature Snapshot
7. State Machine
8. Session Engine
9. Regime Engine
10. Swing / Structure
11. Range Engine
12. Range Strategy
13. Breakout Strategy
14. Trend Strategy
15. Strategy Guardian
16. Risk Engine
17. Portfolio Engine
18. Execution Safety
19. Execution Engine
20. Position Management
21. Trade Logger
22. Backtest
23. Validation
24. AI Dataset
25. AI Probability
26. AI Orchestrator
27. Signal Publisher
28. Analytics
29. Customer Insight
```

---

125. V1 DEFINITION OF DONE

V1 tidak selesai hanya karena EA membuka posisi.

Required:

```
✓ Semua module berjalan
✓ Deterministic strategy
✓ No look-ahead bias
✓ No hidden parameter
✓ Configuration versioning
✓ Feature versioning
✓ Backtest reproducible
✓ Trade log lengkap
✓ Decision audit
✓ Risk calculation validated
✓ Portfolio exposure validated
✓ Execution logic validated
✓ OOS selesai
✓ Walk-forward selesai
✓ Monte Carlo selesai
✓ Stress test selesai
✓ Multi-pair test selesai
```

```
✓ Multi-session analysis selesai
✓ Regime analysis selesai
✓ Forward test selesai
```

126. V2 DEFINITION OF DONE

```
✓ V1 validated
✓ Dataset clean
✓ No leakage
✓ Model versioned
✓ Output schema fixed
✓ Guardian operational
✓ AI timeout handled
✓ AI failure handled
✓ Hard risk immutable
✓ V1 vs V2 comparison complete
✓ OOS AI positive/acceptable
✓ Walk-forward acceptable
✓ Stress acceptable
✓ Model monitoring available
```

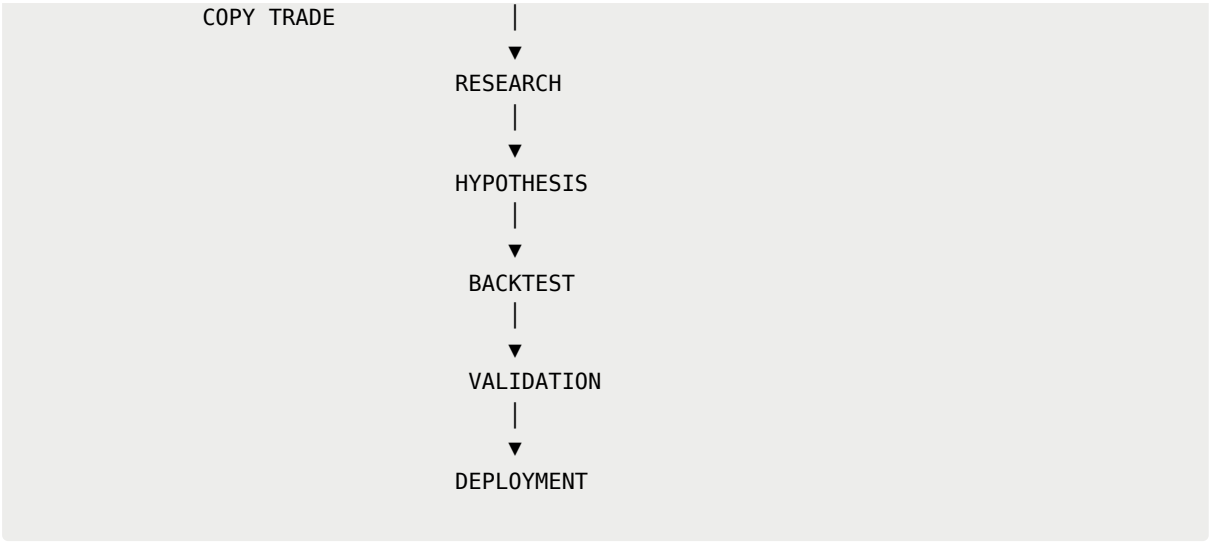
127. ABSOLUTE RULES

Developer tidak boleh melakukan:

```
✗ Hidden parameter
✗ Future data
✗ Look-ahead bias
✗ Martingale
✗ Recovery position sizing
✗ Averaging down for recovery
✗ AI override hard risk
✗ AI direct broker execution
✗ Auto retrain live
✗ Auto deploy model
✗ Silent live parameter change
✗ Change strategy because of one losing period
✗ Remove pair without validation
✗ Different trading decision per signal channel
✗ Agent calculating independent market feature
✗ Research directly modifying production
```

128. FINAL ARCHITECTURE

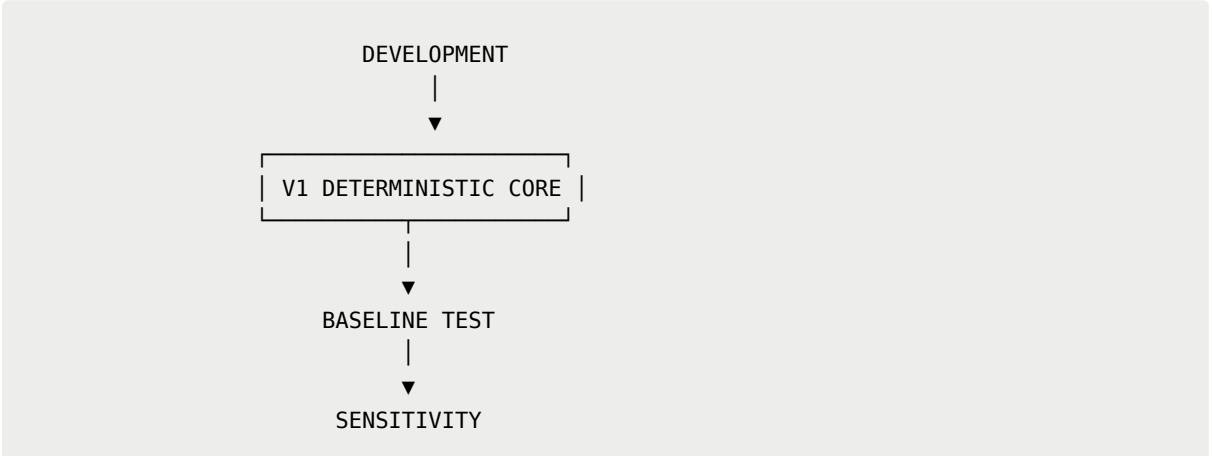


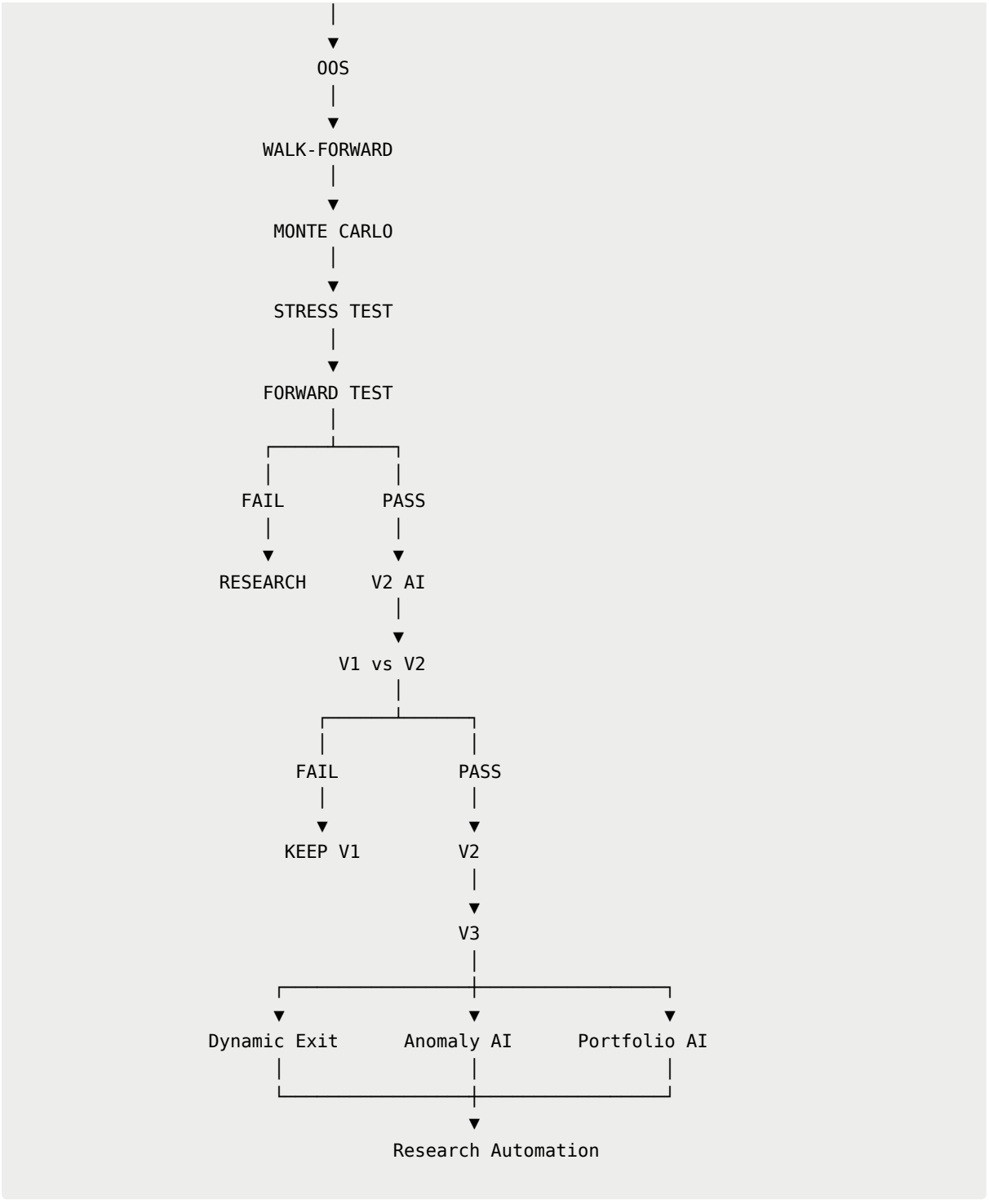


129. FINAL DEVELOPMENT RULE

> **Build V1 as a deterministic, modular, reproducible, auditable trading system. Feature Engine is the single source of truth. Regime Engine determines market state before Strategy Engine generates a candidate. Strategy never executes. Guardian validates deterministic conditions. AI, beginning in V2, provides probability/intelligence only. Risk and Portfolio remain authoritative. Execution remains deterministic and isolated from AI. Every decision must be reconstructable through immutable snapshots and versioned configuration. No look-ahead bias, martingale, hidden parameters, live self-learning, silent parameter changes, or unvalidated production changes are permitted.**

130. V1 → V2 → V3 ROADMAP





STATUS: V1.0 MASTER SPECIFICATION — FROZEN

Dengan revisi ini, saya akan menganggap beberapa hal sebagai **fixed engineering contract**, sementara angka seperti:

```
ADX = 20
Range = 20 candles
Range zone = 20%
Body = 0.5 ATR
SL buffer = 0.2 ATR
Base risk = 0.5%
```

tetap **baseline experimental**, bukan parameter optimal.

Perbedaan pentingnya adalah sekarang kita sudah mengunci hal-hal yang memang **tidak boleh berubah sembarangan**, seperti:

- closed-candle decision timing;
- confirmed swing;
- no future information;
- Feature Engine sebagai single source of truth;
- immutable snapshots;
- module contracts;
- authority hierarchy;
- configuration version;
- deterministic requirement;
- AI failure policy;
- portfolio authority;
- execution separation;
- research/production separation;
- audit trail;
- validation gates.